

Game Architecture Summary - Complete Edition

Core Concept

Twin Peaks-inspired mystery board game combining circular time-based movement, resource management, dynamic event system, and deck-building through character encounters. Players investigate mysteries while managing time, collecting allies, and navigating permanent consequences.

File Structure

Autoload Singletons (Global)

Path: Project Settings → Autoload

GameState.gd

- **Purpose:** Global game progression tracking
- **Variables:**
 - `current_day: int` - Day counter
- **Methods:**
 - `increment_day()` - Advances day, emits signal

SignalBus.gd

- **Purpose:** Global event communication hub
- **Signals:**
 - `player_landed_on_tile(tile_index, hour)`
 - `day_changed(new_day)`
 - `midnight_crossed`
 - `popup_requested(event: Event)`
 - `popup_closed`
 - `board_refresh_requested` - Force visual update

TileEventManager.gd

- **Purpose:** Event placement, recurrence, tracking
- **Variables:**
 - `tile_events: Dictionary` - hour → Event
 - `multi_tile_event_map: Dictionary` - hour → starting hour

- `locked_tiles: Dictionary` - Permanent events (curses/blessings)
- `event_occurrence_count: Dictionary` - `event_id` → times triggered
- `event_last_triggered_day: Dictionary` - `event_id` → last day
- `events_used_today: Dictionary` - Daily usage tracker
- Event pools: `standard_event_pool`, `character_encounter_pool`, `curse_pool`, `blessing_pool`
- **Key Methods:**
 - `setup_events()` - Place events for current day
 - `lock_tile(hour, event)` - Permanently assign event
 - `unlock_tile(hour)` - Remove permanent event
 - `can_event_occur(event)` - Check recurrence/cooldown rules
 - `get_random_curse/blessing/character_encounter()` - Get specific event types

`CharacterDatabase.gd`

- **Purpose:** Character card definitions
- **Variables:**
 - `characters: Dictionary` - `character_id` → Character
- **Methods:**
 - `setup_characters()` - Creates all characters
 - `get_character(id)` - Get specific character
 - `get_random_character()` - Random character

`ResourceManager.gd`

- **Purpose:** Player resources
- **Variables:**
 - `money: int`
 - `insight: int`
 - `time: int`
- **Signals:**
 - `resource_changed(type, new_amount)`
- **Methods:**
 - `add_resource(pickup)` - Add/subtract resources
 - `get_random_resource()` - Random resource pickup

`DeckManager.gd`

- **Purpose:** Card deck management
 - **Variables:**
 - `deck: Array[Card]` - Player's current deck
 - `card_pool: Array[Card]` - All available cards
 - `acquired_card_ids: Array[String]` - Track unique characters
 - **Signals:**
 - `card_added(card)`
 - `card_removed(card)`
 - `deck_updated`
 - **Methods:**
 - `add_card(card)` - Add to deck (prevents duplicate characters)
 - `add_card_by_id(id)` - Add by ID
 - `get_random_card()` - Get random card (respects uniqueness)
 - `get_cards_by_type(type)` - Filter by card type
 - `get_deck_stats()` - Count by type
-

Resource Classes

`Character.gd`

- **Type:** Resource
- **Properties:**
 - `character_name: String`
 - `title: String`
 - `description: String`
 - `portrait: Texture2D`
 - `rarity: String`
 - `insight: int` (stat)
 - `connection: int` (stat)

`Event.gd`

- **Type:** Resource
- **Properties:**
 - `event_id: String`
 - `title: String`

- `description: String`
- `image: Texture2D`
- `event_type: String` - standard, character_encounter, resource, curse, blessing
- `duration_hours: int` - 1-5+ hours
- `recurrence: RecurrenceType` - UNIQUE, LIMITED, UNLIMITED
- `max_occurrences: int` - For LIMITED type
- `cooldown_days: int` - Days between repeats
- `time_windows: Array[Dictionary]` - Multiple time ranges
- `resource_pickup: ResourcePickup` - Reward/penalty
- **Methods:**
 - `can_occur_at_hour(hour)` - Check time restrictions
 - `add_time_window(start, end)` - Add valid time range
 - `is_curse/blessing/permanent()` - Type checks

`ResourcePickup.gd`

- **Type:** Resource
- **Enum:** MONEY, INSIGHT, TIME, ALL
- **Properties:**
 - `resource_type: ResourceType`
 - `display_name: String`
 - `amount: int` - Can be negative
 - `description: String`

`Card.gd`

- **Type:** Resource
- **Enum:** CHARACTER, ITEM, ACTION, RITUAL
- **Properties:**
 - `card_id: String`
 - `card_name: String`
 - `card_type: CardType`
 - `description: String`
 - `portrait: Texture2D`
 - `uses_remaining: int` - -1 = infinite, >0 = limited
- **Methods:**

- `get_type_name()` - Human-readable type
 - `is_permanent()` - Check if infinite uses
 - `use_item()` - Consume one use
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Main Scenes

`Board.tscn` (Node2D)

- **Purpose:** Main game loop
- **Children:**
 - $24 \times$ BoardTile (created at runtime)
 - Player token (created at runtime)
 - CanvasLayer (UI)
 - Roll Die Button
 - View Deck Button
 - PlayerStatsUI
 - EventPopup
 - DeckView

`BoardTile.tscn` (Area2D)

- **Script:** BoardTile.gd
- **Properties:**
 - `hour_value: int` - What hour (0-23) this tile represents
- **Children (created at runtime):**
 - Polygon2D - Wedge shape
 - CollisionPolygon2D - Click detection
 - Line2D - Border

`Player.tscn` (Node2D)

- Visual player token
- No script needed

`EventPopup.tscn` (CanvasLayer)

- **Script:** EventPopup.gd
- Displays event title, description, image

- Handles resource collection
- Adds cards to deck for character encounters

PlayerStatsUI.tscn (**PanelContainer**)

- **Script:** PlayerStatsUI.gd
- Shows Money, Insight, Time with animated updates
- Shows current day

DeckView.tscn (**CanvasLayer**)

- **Script:** DeckView.gd
- Full-screen overlay
- Left panel: Deck stats (total, by type)
- Right panel: Scrollable card list grouped by type
- Hover effects on cards

CharacterCard.tscn (**PanelContainer**)

- **Script:** CharacterCard.gd
 - Visual card display (not currently used in deck view)
 - For future: hand display, card selection
-

Core Systems

1. Board & Time System

Board Layout:

- 24 tiles in circle, each = 1 hour (00-23)
- **Midnight (00)** at bottom, **Noon (12)** at top
- Player starts at **06:00** (morning)
- Tiles rotate clockwise with gradient day/night colors

Color Gradient:

Night (21-04): Deep blue

Dawn (05-07): Orange/pink transition

Day (08-16): Yellow to purple

Dusk (17-20): Purple to blue

Tile Coordinates:

- `tile_index` - Physical position (0-23, 0 at top going clockwise)
 - `hour_value` - Time displayed (0-23)
 - Conversion: $\text{hour} = (\text{tile_index} + \text{FIRST_TILE_HOUR}) \% 24$
-

2. Movement System

Dice Roll Movement:

1. Roll 1-6
2. Calculate destination: $(\text{current} + \text{roll}) \% 24$
3. Generate clockwise path
4. Animate tile-by-tile (0.4s per tile)
5. If crossing midnight (23→00): pause, show day popup, continue
6. If land on event: show popup
7. If multi-tile event: auto-move to end tile after popup

Movement Blocking:

- `is_moving` flag prevents clicks during animation
- Blocks during: dice roll movement, midnight pause, event end-tile movement

Day Change:

- Triggers when crossing from hour 23 → hour 00
 - Animation pauses at midnight
 - Day counter increments
 - Events reshuffle
 - Day popup shows
 - Animation resumes after popup close
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3. Event System

Event Types:

- **Standard** - Narrative events (gold borders)
- **Character Encounter** - Adds card to deck (gold borders)
- **Resource** - Gain/lose resources (orange borders)
- **Curse** - Permanent negative tile (red borders, locked)
- **Blessing** - Permanent positive tile (green borders, locked)

Event Duration:

- Events span 1-5+ tiles (hours)
- Multi-tile events have unified color
- Visual dividers separate events
- Player auto-moves to end tile after event

Recurrence System:

UNIQUE: Only once in entire game

LIMITED: X times total (max_occurrences)

UNLIMITED: Can repeat infinitely

+ cooldown_days: Minimum days between occurrences

+ events_used_today: Can't repeat same day

Time Restrictions:

```
gdsript
```

```
// Single window
```

```
available_start_hour: 8
```

```
available_end_hour: 17 // Event only 8-17
```

```
// Multiple windows
```

```
time_windows: [
```

```
  {start: 19, end: 23}, // Evening OR
```

```
  {start: 1, end: 5} // Early morning
```

```
]
```

Event Placement (Each Day):

1. Clear non-locked events
2. Place locked tiles (curses/blessings)

3. Place lunch at hour 12 (if unlocked)
4. Random events (2-4): Check recurrence, cooldown, time windows, no overlap
5. Random resources (3-5): Same checks
6. Highlight all event tiles
7. Hour 00 (midnight) never gets events

Locked Tiles:

- Curses/blessings permanently occupy tiles
 - Persist across day changes
 - Exempt from random assignment
 - Can be unlocked programmatically
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4. Resource System

Three Resources:

- 💰 **Money** - Currency (future: purchases, bribes)
- 🔮 **Insight** - Investigation resource (used for rituals)
- ⌚ **Time** - Investigation deadline extension

Resource Sources:

- Resource tiles (orange, random placement)
- Event rewards (any event can grant)
- Blessings (permanent positive tiles)
- Curses (permanent negative tiles, subtract resources)
- Resources can be negative (curses), floor at 0

UI Feedback:

- Animated value changes (scale pulse + color flash)
 - Yellow flash on gain
 - Updates on resource_changed signal
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5. Card Deck System

Card Types:

CHARACTER - People you meet (unique, permanent)

ITEM - Tools with limited/unlimited uses

ACTION - Activities (permanent)

RITUAL - Complex actions requiring resources/items (permanent)

Uniqueness Rules:

- **Characters:** Can only be acquired once (tracked by card_id)
- **Items/Actions/Rituals:** Can have duplicates

Starting Deck:

1. "You" (Character) - The player detective
2. "Notebook" (Item) - Evidence recording
3. "Talk" (Action) - Communication

Deck View UI:

- Full-screen overlay (doesn't change scene)
- Left panel: Stats (total, by type), Synergies (future)
- Right panel: Scrollable card list grouped by type
- Hover effect: Yellow glow
- Return to Board button

Card Acquisition:

- Character encounter events add random cards
- `get_random_card()` respects character uniqueness
- Cards immediately added to deck, no choice (narrative-driven)

Future Systems (Not Implemented):

- Card synergies
- Ritual requirements (needs specific cards/resources)
- Item consumption
- Character abilities

Key Constants (Board.gd)

```
// Board structure
TILE_COUNT = 24
RADIUS = 300
INNER_RADIUS = 220
FIRST_TILE_HOUR = 0 // Midnight at top-left
MOVE_DURATION_PER_TILE = 0.4

// Visual colors - Event tiles
EVENT_HIGHLIGHT_COLOR = Color(1.4, 1.3, 1.1) // Gold glow
RESOURCE_HIGHLIGHT_COLOR = Color(1.5, 1.2, 0.8) // Orange glow
CURSE_TILE_COLOR = Color(1.4, 0.9, 0.9) // Red tint
BLESSING_TILE_COLOR = Color(0.9, 1.4, 1.0) // Green tint
LOCKED_TILE_COLOR = Color(1.3, 1.0, 1.0) // Legacy red tint
TILE_HOVER_COLOR = Color(1.0, 1.0, 1.0, 1.0) // White

// Visual colors - Dividers
EVENT_DIVIDER_COLOR = Color(1.0, 0.8, 0.0, 0.9) // Gold
RESOURCE_DIVIDER_COLOR = Color(1.0, 0.5, 0.0, 0.9) // Orange
CURSE_DIVIDER_COLOR = Color(1.0, 0.0, 0.0, 1.0) // Red
BLESSING_DIVIDER_COLOR = Color(0.0, 1.0, 0.3, 1.0) // Green
```

Data Flow Examples

Turn Sequence

1. Player clicks "Roll Die"
2. Roll 1-6, calculate destination
3. Generate clockwise path
4. **Check for midnight crossing:**
 - If path crosses hour 00: Split movement
 - Move to midnight → pause → day popup → continue
5. Animate tile-by-tile to destination
6. **Landing on event:**
 - Check if tile has event
 - Show event popup
 - Collect resources if resource event
 - Add card if character encounter
 - **Multi-tile event:** Auto-move to end tile after popup
7. Re-enable roll button

Day Change Flow

1. Player animation reaches midnight (hour 00)
2. Movement pauses (`is_moving = true` stays)
3. `GameState.increment_day()`
4. `TileEventManager.setup_events()` - Reshuffle events
5. Clear and re-highlight event tiles
6. Show day change popup
7. User closes popup → `SignalBus.popup_closed.emit()`
8. Movement resumes to final destination
9. Normal landing sequence continues

Event Placement Algorithm

FOR each new day:

1. Clear `tile_events` and `multi_tile_event_map`
2. Keep `locked_tiles` (persistent)
3. Clear `events_used_today`
4. Place locked tiles (curses/blessings)
5. IF hour 12 not locked:
Place lunch event (2 hours)
6. FOR 2-4 random standard events:
 - Pick random from `event_pool`
 - Check `can_event_occur()`:
 - * Not used today
 - * Recurrence allows (UNIQUE/LIMITED/UNLIMITED)
 - * Cooldown satisfied
 - Get available hours:
 - * Respect `time_windows`
 - * No overlap with existing events
 - * Event fits before midnight
 - Place if possible
 - Mark as used today
7. FOR 3-5 random resources:
 - Similar placement logic
 - Resources always 1 hour
8. Highlight all event tiles with colored borders

Card Acquisition

1. Player lands on character_encounter event
 2. EventPopup shows event
 3. `DeckManager.get_random_card()` called
 4. Filters out already-acquired characters
 5. Returns random available card
 6. `DeckManager.add_card()` adds to deck
 7. Popup shows "[Added to deck: Card Name]"
 8. `deck_updated` signal refreshes UI
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Example Code Snippets

Creating Custom Event with Time Windows

```
gdscript

var night_patrol = Event.new(
    "night_patrol",
    "Night Patrol",
    "You patrol the dark streets.",
    null,
    "standard",
    3, // 3 hours duration
    Event.RecurrenceType.UNLIMITED,
    -1,
    1 // 1 day cooldown
)
// Add multiple time windows
night_patrol.add_time_window(20, 23) // Evening
night_patrol.add_time_window(1, 4) // Early morning
TileEventManager.standard_event_pool.append(night_patrol)
```

Locking a Curse

```
gdscript

var curse = TileEventManager.get_random_curse()
TileEventManager.lock_tile(15, curse)
SignalBus.board_refresh_requested.emit() // Force visual update
```

Adding Resource to Event

gdsript

```
var mystery_event = Event.new(...)
mystery_event.resource_pickup = ResourcePickup.new(
  ResourcePickup.ResourceType.INSIGHT,
  "Insight",
  2,
  "You piece together the clues."
)
```

Creating Custom Card

gdsript

```
var flashlight = Card.new(
  "item_flashlight",
  "Flashlight",
  Card.CardType.ITEM,
  "Illuminates dark places.",
  null,
  3 // 3 uses
)
DeckManager.card_pool.append(flashlight)
```

Current Game Loop

1. **Start:** Day 1, Hour 06:00, 3 starting cards
 2. **Roll die** → move clockwise
 3. **Land on tiles** → trigger events/resources/cards
 4. **Cross midnight** → new day, events reshuffle
 5. **Resources accumulate** → displayed in UI
 6. **Cards collected** → viewable in deck
 7. **Permanent tiles** → curses/blessings persist
 8. **Repeat**
-

Not Yet Implemented

Planned Features

- **Card mechanics:** Synergies, abilities, active use

- **Rituals:** Require specific cards + resources
- **Item consumption:** Use limited-use items
- **Win condition:** Solve mystery before time runs out
- **Quest system:** Long-term objectives
- **Combat encounters:** Transition to battle scene
- **Save/load:** Persist game state
- **Character portraits:** Visual art for cards/events
- **Sound/music:** Audio atmosphere
- **Narrative branches:** Story choices affecting outcome
- **Card selection:** Choose which cards to keep (vs current auto-add)

Architecture Ready For

- Event system supports scene transitions (combat, investigation)
 - Card types support complex interactions (rituals need items/characters)
 - Resource system expandable (new resource types easy to add)
 - Signal bus allows disconnected systems to communicate
 - Locked tiles enable permanent world changes from player actions
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Technical Notes

Performance Considerations

- All tiles created at startup (24 nodes)
- Events refreshed only on day change
- Card list rebuilt only when deck changes
- Tween animations use single tween per movement

Godot 4 Specific

- Uses typed GDScript (`Array[Card]`, etc.)
- Resource-based data (Event, Card, Character, ResourcePickup)
- Signal-based communication (SignalBus pattern)
- CanvasLayer for UI overlays
- `Control.SIZE_EXPAND_FILL` for responsive UI

Common Gotchas

- `is_moving` must be set/cleared properly to prevent input spam
- Character uniqueness tracked separately from deck
- Time windows can wrap (21-02) but avoid midnight (00)
- Multi-tile events update `player_tile_index` after animation
- Deck view needs manual size forcing due to Godot 4.6 layout quirks
- Event `duplicate(true)` needed to avoid shared resource instances